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Explainable extreme gradient boosting as a machine learning tool for discrimination of the geographical origin of chili peppers using laser ablation-inductively coupled plasma mass spectrometry, X-ray fluorescence, and near-infrared spectroscopy

Seongsoo Jeong [a,1], Yong-kyoung Kim [b,1], Suel Hye Hur [b], Hyojoo Bang [a], HoJin Kim [b,**], Hoeil Chung [a,*]

[a] Department of Chemistry and Research Institute for Convergence of Basic Science, Hanyang University, Seoul 04763, Republic of Korea
[b] National Agricultural Products Quality Management Service, Gimcheon, 39660, Republic of Korea

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ABSTRACT

The spectroscopic discrimination of chili pepper samples according to geographical origin was executed using analytical techniques coupled with machine learning. First, laser ablation-inductively coupled plasma mass spectrometry (LA-ICP-MS), X-ray fluorescence (XRF), and near-infrared (NIR) spectroscopy were chosen for simple and rapid sample measurements. Second, to secure discrimination accuracy, eXtreme Gradient Boosting (XGBoost), a tree-based ensemble technique, was adopted as a potential classifier. Also, for explainable machine learning modeling, SHaply Additive exPlanation (SHAP) values of employed variables were calculated to assess how they contribute to the discrimination. The use of XGBoost improved discrimination accuracies in all three measurements compared to *k*-nearest neighbor (*k*-NN), support vector machine (SVM), and partial least squares-discriminant analysis (PLS-DA). The accuracy was 96.2 % using the LA-ICP-MS data. When the XRF and NIR data were combined, the accuracy improved to 97.5 %. The accuracy improvement was attributed to the combination of complementary atomic and molecular spectroscopic signatures of the samples.

1. Introduction

Recently, international trading of diverse agricultural products has been rapidly growing to secure stable food supplies and satisfy the diversified tastes of consumers. Consumers believe that the country of origin is linked to the quality of food [1], which has led to an increase in unscrupulous traders taking advantage of this by incorrect origin marking (labeling). Especially, domestic products are regarded as more flavorful than imported products and are generally more expensive [2]. Therefore, the demand for authentication of agricultural products according to geographical origin has increased and fast spectroscopic methods with minimal or no sample pretreatment are desirable to handle large number of samples in a real field.

Chili peppers are an economically important crop grown worldwide for their culinary and nutritional value [3], containing amino acids such as aspartic and glutamic acids, and nutrients such as vitamin C, vitamin A, calcium, and magnesium [4]. Furthermore, it is a popular food and important condiment in Korean foods (K-Foods) such as Kimchi and Ddeokbokki (spicy rice cake). In general, it is not only used directly in cooking but is also often processed or ground after drying. In this case, it is difficult to distinguish the geographical origin of dried chili pepper powder by its ground form. So, sophisticated analytical methods are required for this challenging authentication and the combination of multiple analytical data may be required to secure accuracy. Under this situation, an analysis tool effectively utilizing the multiple data expects to be in demand.

In this sense, laser ablation-inductively coupled plasma mass spectrometry (LA-ICP-MS) and X-ray fluorescence (XRF) are preferred candidates in atomic spectroscopy. As previously reported, elemental compositions of the samples vary by environmental (soil, water, and

* Co-corresponding author.
** Co-corresponding author.
E-mail addresses: rex7878@korea.kr (H. Kim), hoeil@hanyang.ac.kr (H. Chung).
[1] The first two authors contributed equally to this work.

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climate) and production (fertilizers and inorganic pesticides/herbicides) conditions. Therefore, atomic spectroscopy-based authentications of paprika powder (Pimentón de la Vera), extra virgin olive oil, cheese, coffee bean, and salt have been reported [5–8]. Also, compositions of organic components in samples could also differ according to their origins, Raman and near-infrared (NIR) spectroscopy which enable to feature out the dissimilar compositions are alternatively choices for authentication. Meanwhile, since many agricultural products are colored and potentially generate fluorescence obscuring Raman peaks of sample, NIR spectroscopy has been more frequently adopted for authentication of the geographical origins of various agricultural products, such as olive oil [9] and soybeans [10].

In addition to the inherent spectral selectivity and sensitivity provided by each of the mentioned fast spectroscopic techniques, a potential way to improve accuracy further is the adoption of a classification tool (classifier) that could more effectively utilize input variables, which are intensity and concentration of element peaks, and absorbance of NIR bands, for discriminant analysis. For this purpose, machine learning classification tools such as *k*-Nearest Neighbor (*k*-NN) and support vector machine (SVM) have been adopted to identify the geographical origins of millet and white asparagus products using their NIR data [11, 12], and pork products using their ICP-MS data [13]. More recently, eXtreme Gradient Boosting (XGBoost) has been utilized for the authentication of turmeric powders using their NIR data [14] and Australian Cabernet Sauvignon wines using their ICP-MS data [15]. Meanwhile, the interpretations of each XGBoost model to the corresponding authentication were rather deficient.

XGBoost [16], a tree-based ensemble technique, has gained considerable attention nowadays and could be a viable choice for discriminant analysis. In XGBoost, a strong learner is generated by combining numerous weak learners from earlier iterations. This step is executed by expanding loss and minimizing the residual using gradient descent while fitting decision trees at each step. As the use of XGBoost for atomic/vibrational spectroscopic analysis in an early growing stage as described above, it is worthwhile to explore XGBoost as a potential discriminant tool for authentications of agricultural products, especially when diverse types of analytical data are separately employed or combined.

To examine the potential of XGBoost, discrimination of imported and domestic chili pepper samples using LA-ICP-MS, XRF, and NIR spectroscopy was readily executed in this study. Discrimination accuracies of each method were obtained using XGBoost and compared with those based on *k*-NN, SVM, and partial least squares-discriminant analysis (PLS-DA) [17]. Since XGBoost is also based on black box-like modeling, explainable artificial intelligence (XAI) approaches (intended to rationalize the relationship between inputs and outputs) are beneficial to support subsequent discrimination results. For this purpose, the SHaply Additive exPlanation (SHAP) method [18] was adopted as a metric of feature attribution value based on an integration of principles from game theory. It integrates consistent local interpretations into a global understanding of the model. Thus, SHAP values of the input variables were used to examine their importance and individual contributions to the discrimination.

Further, the combination of information on elemental and organic compositions of the samples could improve discrimination accuracy, since the two are potentially synergetic for recognizing differences in samples. For this purpose, the principal component (PC) scores of NIR spectra were merged with the intensities and concentrations of element peaks acquired either by LA-ICP-MS or XRF for discrimination, and the resulting accuracies were compared with those using data from a single spectroscopic method. Again, SHAP values of the chosen element peaks and the PC scores were examined to assess the contributions of each variable to overall discrimination.

2. Materials and methods

2.1. Preparation of chili pepper and pure component samples

In this study, 101 chili pepper samples composed of 50 imported and 51 domestic samples were employed. A detailed description of the sample set can be found in a previous report [19]. All samples were collected by the National Agricultural Products Quality Management Service (NAQS) in Gimcheon, Korea, an official governmental organization that verifies the country-origin of agricultural products. Briefly, diverse domestic products were obtained from local food stores in all 8 provinces in Korea. The imported samples mainly originated from China and Vietnam. All samples were in powder form. Each powder sample was passed through a 0.5 mm filter, dried in a hot air dryer (40 °C) for 3 h, and measured using LA-ICP-MS, XRF, and NIR instrumentations.

The pure components used to investigate NIR spectral feature of the chili pepper samples were purchased from common suppliers. Ascorbic acid, capsaicin, cellulose, retinol, and thiamine were purchased from Sigma-Aldrich (USA). Glutamic acid, proline, isoleucine, and linoleic acid were obtained from Tokyo Chemical Industry (Japan). Leucine, phenylalanine, and tryptophan were purchased from Junsei Chemical (Japan).

2.2. LA-ICP-MS measurement of chili pepper samples

Chili pepper sample pellets (2–3 mm thickness and 13 mm in diameter) prepared from powder samples were employed for LA-ICP-MS measurements. Approximately 2 g of each powder sample was pressed at 10 ton/cm^2 with a PP35 pellet presser (Retsch, Haan, Germany) for pellet preparation. A femtosecond (fs) LA-ICP-MS instrument composed of a J200 fsLA (Applied Spectra, USA) and an Agilent 7900 ICP-MS (Agilent, USA) was employed to measure the pellets. The optimized parameters used for LA-ICP-MS measurement [19], comprising ablation mode, scanning rate, spot size, laser fluence, repetition rate, laser energy, carrier flow, pulse length, and wavelength, are summarized in Table S1 in the Supplementary Information. The RF power, RF match, and He flow in the ICP-MS measurement were set to 1600 W, 1.40 V, and 2.0 mL/min, respectively. All measurements were conducted inside an ablation chamber (100 mm × 100 mm).

Multi-element analysis (including 61 elements: ^{9}Be, ^{23}Na, ^{24}Mg, ^{27}Al, ^{31}P, ^{34}S, ^{39}K, ^{43}Ca, ^{44}Ca, ^{48}Ti, ^{51}V, ^{52}Cr, ^{55}Mn, ^{56}Fe, ^{59}Co, ^{61}Ni, ^{63}Cu, ^{66}Zn, ^{71}Ga, ^{72}Ge, ^{75}As, ^{78}Se, ^{88}Sr, ^{90}Zr, ^{93}Nb, ^{95}Mo, ^{101}R, ^{103}Rh, ^{105}Pd, ^{107}Ag, ^{111}Cd, ^{118}Sn, ^{121}Sb, ^{125}Te, ^{133}Cs, ^{137}Ba, ^{139}La, ^{140}Ce, ^{141}Pr, ^{146}Nd, ^{147}Sm, ^{153}Eu, ^{157}Gd, ^{163}Dy, ^{165}Ho, ^{166}Er, ^{169}Tm, ^{172}Yb, ^{175}Lu, ^{178}Hf, ^{181}Ta, ^{182}W, ^{185}Re, ^{189}Os, ^{193}Ir, ^{195}Pt, ^{197}Au, ^{205}Tl, ^{208}Pb, ^{232}Th, and ^{238}U) was executed for each sample with four single-line scanning laser ablation modes performed at three locations on the pellet. Among them, 28 elements with unacceptable sensitivity by matrix effect and inferior reproducibility were excluded, and the remaining 33 elements (refer to Table 1) were adopted for subsequent analysis. Quantification of elements using LA-ICP-MS is challenging to perform matrix matching correction because it requires solid CRMs with various concentrations of the same matrix. Therefore, the normalized counts per second (cps) values were used for the discriminant analysis rather than the individual element quantification. Also, 13C [20] and NIST 612 (Synthetic glass CRM, NIST, Gaithersburg, USA) were used as internal standard (IS) and to reduce errors, respectively, to ensure the reproducibility of results between analyses.

2.3. XRF measurement of chili pepper samples

For XRF measurements, 5 g of each powder sample was placed in a sample cup (Chemplex, 30.7 mm diameter × 22.9 mm height, Palm City, FL, USA). An energy-dispersive XRF spectrometer (S2 Ranger, Bruker AXS GmbH, Karlsruhe, Germany) was employed, and a total of 19 elements (Al, Si, P, S, Cl, K, Fe, Cu, Zn, Se, Br, Rb, Sr, Ag, Sn, Hg, Ti, Pb, and

Table 1
The *t*-test results for peak intensity and element concentration difference between the imported and domestic chili pepper samples in the LA-ICP-MS and XRF measurements, respectively. The elements yielding *p*-values smaller than 0.05 (95 % CI)), 0.01 (99.0 % CI), 0.001 (99.9 % CI), or 0.0001 (99.99 % CI) are separately listed. 'S' and 'NS' respectively stand for 'significant' and 'not significant' based on *t*-tests.

t-Test result	LA-ICP-MS measurement							
S (99.99 % CI)	^{27}Al	^{51}V	^{63}Cu	^{66}Zn	^{71}Ga	^{88}Sr	^{111}Cd	^{133}Cs
S (99.9 % CI)	^{23}Na							
S (99.0 % CI)	^{93}Nb	^{95}Mo	^{181}Ta	^{208}Pb	^{238}U			
S (95.0 % CI)	^{48}Ti	^{137}Ba	^{140}Ce	^{141}Pr	^{146}Nd	^{178}Hf	^{205}Tl	
NS	^{24}Mg	^{31}P	^{34}S	^{39}K	^{43}Ca	^{44}Ca	^{52}Cr	^{55}Mn
	^{56}Fe	^{59}Co	^{61}Ni	^{139}La				
XRF measurement								
S (99.99 % CI)	Cu	Zn	Rb					
S (99.9 % CI)	Cl							
S (99.0 % CI)								
S (95.0 % CI)	Br	Ag						
NS	Al	Si	P	S	K	Fe	Sn	

Bi) was measured under the optimized conditions (refer to Table S2 in the Supplementary Information). Here, 13 elements (refer to Table 1) with acceptable sensitivity and reproducibility were chosen for subsequent analysis. A standardless fundamental parameters (SLFP) approach was adopted to analyze the measured element peaks. The SFLP method is typically used for qualitative and quantitative analyses of element peaks when there is no available standard material with the same matrix composition as the sample [21,22]. Using the Spectra EDX program (S2 Ranger, Bruker), the relative content ratio of the elements was calculated by SLFP and used for following discriminant analysis.

2.4. NIR measurement of chili pepper samples

Diffuse reflectance NIR spectra (6050–4000 cm^{-1} range, resolution: 8 cm^{-1}, 128 scans) of the chili pepper samples were collected using an MB3600 FT-NIR spectrometer (ABB Analytical Measurements, Quebec, Canada) equipped with a tungsten-halogen source and an InGaAs detector. Each powder sample was transferred into a glass vial (diameter: 20 mm, height: 50 mm) that was revolved during spectral acquisition (3 revolutions per minute). Teflon powder in a glass vial was used to measure the background. NIR spectra of the pure components (listed in Section 2.1) were acquired using the same spectral acquisition setting. All raw NIR spectra of the samples were treated with multiplicative scatter correction (MSC) to reduce unwanted spectral variations caused by scattering [23].

2.5. Execution of Welch's t-test, k-NN, SVM, PLS-DA and XGBoost

Welch's *t*-test was employed to evaluate statistical significance of the differences in peak intensities and concentrations of elements between imported and domestic samples. Welch's *t*-test was executed using Python (version 3.9.16, Python Software Foundation) with the Scipy package [24]. *k*-NN, SVM, PLS-DA, and XGBoost [16] were used as the classifiers. GridsearchCV for *k*-NN, PLS-DA, and SVM, and Optuna for XGBoost were employed to find optimal hyperparameters in models. Optuna [25], a hyperparameter optimization framework that quickly and simply sets hyperparameters for machine learning methods, could be a suitable choice for XGBoost requiring numerous and complex parameters. The parameter candidates and hyperparameters employed for the analysis of LA-ICP-MS, XRF, and NIR data are summarized in the Supplementary Information (Tables S3 and S4).

For evaluation of discrimination performance, 80 and 20 % of the whole samples were randomly assigned as a training and validation set, respectively. A model was developed using the training set and classes of samples in the validation set were determined using the developed model. The random sample selection for the training and validation sets was executed to maintain the same ratio of 50 imported and 51 domestic samples in the whole set. This procedure was repeated 100 times using a randomly shuffled dataset in each cross validation, and the resulting mean accuracy, specificity, and sensitivity were calculated. *k*-NN, SVM, PLS-DA, and GridSearchCV were executed using the Scikit-learn package [26].

3. Results and discussion

3.1. Intensities of element peaks and elemental concentrations of chili pepper samples

Fig. 1 (a) and (b) show the average intensities (equally counts per second) and concentrations of elements measured from all samples using LA-ICP-MS (33 elements) and XRF (13 elements), respectively. The elements in each plot were displayed with a decreasing order of intensity (counts per second) and element concentration (mg kg^{-1}). The insets highlight the elements with low intensities in each measurement. In the LA-ICP-MS measurement, the K peak was strongest, followed by the Mg peak. Next, the intensities of Na, Fe, and P peaks were weaker than that of Mg peak, and many other elements with smaller peak intensities were observed. In the XRF measurement, the element concentration of K was again highest, and element concentrations of P, S, and Cl were notable. The element concentrations of the remaining 8 elements were substantially lower than those of the above elements. Based on both measurements, K, Na, and Mg were abundant in the chili pepper samples. Meanwhile, nonmetallic elements such as Cl, P, and S were apparent in the XRF measurements.

The intensities and element concentrations between the two groups need to be dissimilar for discrimination of imported and domestic samples. To statistically evaluate the differences in peak intensities between the two groups, *t*-test was performed, and the resulting *p*-values were calculated as summarized in Table 1. The elements yielding *p*-values smaller than 0.05 (95 % confidence interval (CI)), 0.01 (99.0 % CI), 0.001 (99.9 % CI), or 0.0001 (99.99 % CI) are listed separately. The intensities and element concentrations of all the imported and domestic samples for each element are displayed in Fig. S1 (refer to the Supplementary Information). The corresponding averages and standard deviations are also designated in Fig. S1. The *p*-values smaller than 0.05, 0.01, 0.001, or 0.0001 are designated by *, **, ***, or ****, respectively.

In the LA-ICP-MS measurement, the intensities of 8 elements (^{27}Al, ^{51}V, ^{63}Cu, ^{66}Zn, ^{71}Ga, ^{88}Sr, ^{111}Cd, and ^{133}Cs) between the two groups were different within a 99.99 % CI. The intensities of ^{23}Na and 5 other elements (^{93}Nb, ^{95}Mo, ^{181}Ta, ^{208}Pb, and ^{238}U) were different with CIs of 99.9 and 99.0 %, respectively. Intensities of another 7 elements (^{48}Ti, ^{137}Ba, ^{140}Ce, ^{141}Pr, ^{146}Nd, ^{178}Hf, and ^{205}Tl) were different with 95.0 % CI. The intensity differences of the remaining 12 elements between the two groups were not significant. In the XRF measurements, the element

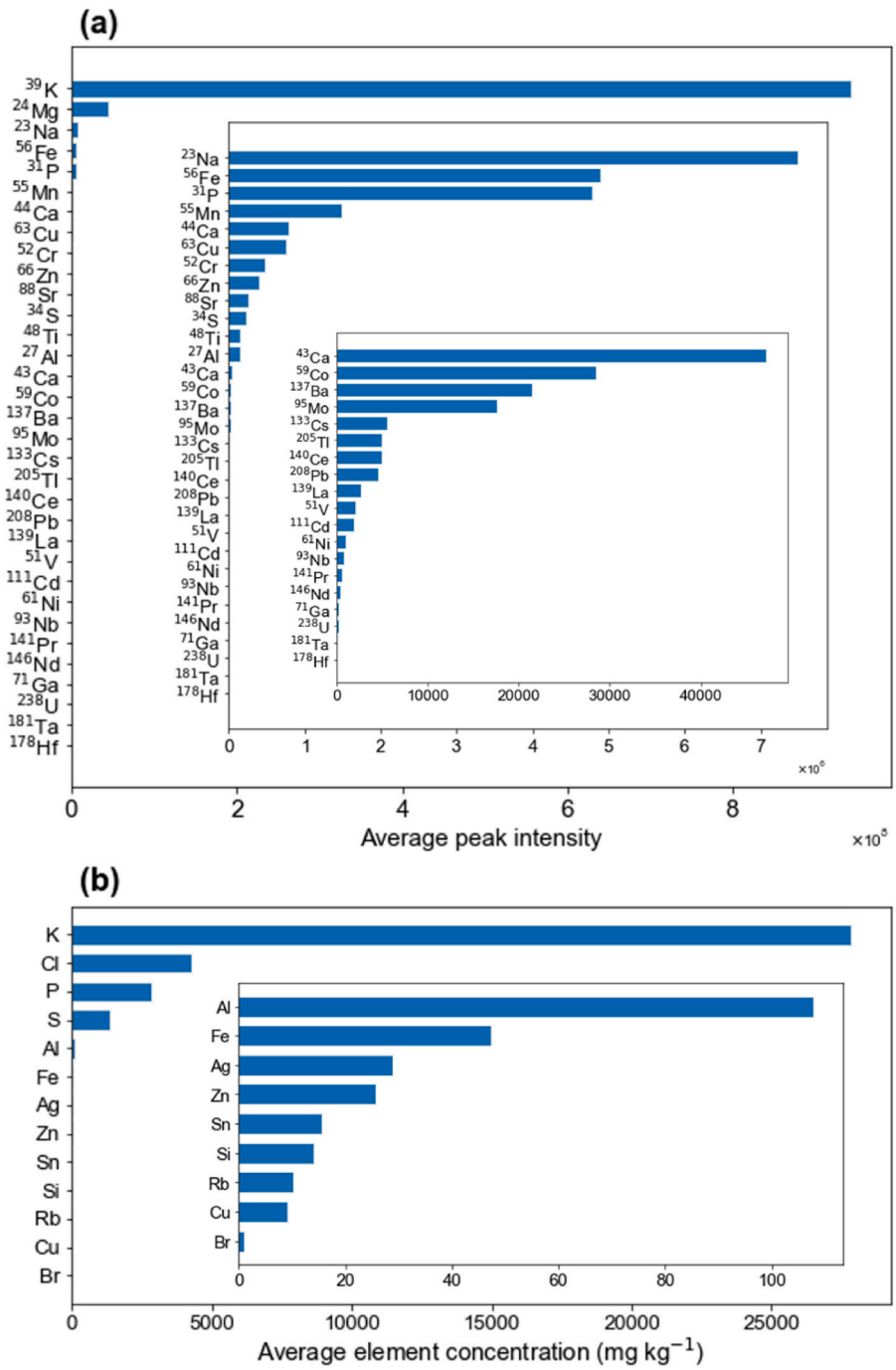

Fig. 1. The average intensities and element concentrations of element peaks in LA-ICP-MS (a) and XRF (b) measurements. The insets highlight the elements with low intensities in each measurement.

concentration differences of Cu, Zn, and Rb between the two groups were most significant with a 99.99 % CI. The element concentrations of Cl and the 2 elements (Br and Ag) were different with CIs of 99.9 and 95.0 %, respectively. The element concentration differences of the remaining 8 elements were not statistically significant.

3.2. Discrimination of imported and domestic samples using elemental peaks

In the LA-ICP-MS and XRF measurements, the elements with larger intensity difference should be more informative for discrimination. When the peak intensity and element concentration distributions of imported and domestic samples were examined (refer to Fig. S1) for the 8 elements with the greatest intensity difference (99.99 % CI, Table 1), they still considerably overlapped with each other. Thus, no elements enabled clear two-group separation, i.e., no so-called marker elements. In this situation, the use of multiple elements for discriminant analysis was a clear choice to ensure accuracy. Therefore, using the peak intensities and element concentrations of all measured elements, the discrimination of imported and domestic samples was performed using *k*-NN, SVM, PLS-DA, and XGBoost, and subsequent results were compared as shown in Table 2. Using all 33 elements from the LA-ICP-MS measurement, the accuracy using XGBoost (96.2 %) was substantially higher than that using *k*-NN (89.6 %), SVM (91.8 %), and PLS-DA (93.2 %). In the XRF measurement utilizing all 13 elements, the accuracies were similar (approximately 90 %) for SVM and XGBoost. Based

Table 2
Accuracies, specificities, and sensitivities for discrimination of the imported and domestic chili pepper samples when different input variables are employed for *k*-NN, SVM, PLS-DA, and XGBoost.

Analytical Method	Input	Classifier	Accuracy (%)	Specificity (%)	Sensitivity (%)
LA-ICP-MS	All 33 elements	*k*-NN	89.6 ± 2.0	92.8 ± 2.3	86.5 ± 3.2
		SVM	91.8 ± 1.8	93.3 ± 2.8	90.2 ± 3.3
		PLS-DA	93.2 ± 1.7	92.1 ± 1.6	94.2 ± 2.9
		XGBoost	96.2 ± 1.0	95.4 ± 1.2	96.9 ± 1.6
XRF	All 13 elements	*k*-NN	81.1 ± 2.9	80.6 ± 4.1	81.6 ± 4.0
		SVM	90.0 ± 2.3	89.6 ± 3.8	90.3 ± 3.1
		PLS-DA	89.2 ± 1.5	92.2 ± 1.9	86.4 ± 2.1
		XGBoost	90.5 ± 1.8	91.4 ± 2.9	89.5 ± 1.8
LA-ICP-MS & XRF	All 33 elements & 13 elements	XGBoost	96.8 ± 1.0	97.0 ± 1.2	96.5 ± 1.7
NIR	5 PC scores	*k*-NN	93.5 ± 1.2	95.7 ± 1.9	91.3 ± 1.3
		SVM	92.8 ± 1.0	95.1 ± 1.4	90.4 ± 1.5
		PLS-DA	92.7 ± 1.2	92.7 ± 1.6	92.5 ± 1.7
		XGBoost	94.1 ± 1.6	94.1 ± 2.4	94.0 ± 1.6
NIR & LA-ICP-MS	5 PC scores & 33 elements	XGBoost	97.2 ± 1.0	97.2 ± 1.3	97.2 ± 1.4
NIR & XRF	5 PC scores & 13 elements	XGBoost	97.5 ± 0.8	97.3 ± 1.2	97.6 ± 1.3

on these observations, XGBoost tends to outperform SVM for the discrimination when a larger number of variables is involved. Meanwhile, the *k*-NN-based accuracies were inferior to both methods (SVM and XGBoost).

To gain additional insight into the XGBoost-based discriminant analysis, the contributions of elements to the discrimination were assessed based on the mean absolute SHAP values. These values indicate how input features (variables) affect the output (discrimination) of a model. The tree SHAP method has consistent values depending on the shape of the model, while the conventional feature importance methods for XGBoost, Gain and Weight do not [27]. Using a SHAP value, it is possible to determine whether a feature has a positive or negative contribution to the output. In this study, the domestic and imported samples were designated as '0' and '1', respectively, for XGBoost modeling, thereby resulting in respective negative and positive SHAP values. The mean absolute SHAP values were used to investigate their importance for the output (discrimination result) of the XGBoost model (see Fig. 2).

Fig. 2 (a) shows the elements with the top 10 highest mean absolute SHAP values in the LA-ICP-MS measurement. The corresponding mean absolute SHAP values are indicated inside each box. Among the 10 elements, ^{133}Cs shows the highest mean absolute SHAP value, indicating that it contributes most significantly to the discrimination. The subsequent mean SHAP values decreased in the order of ^{88}Sr, ^{66}Zn, ^{51}V, ^{27}Al, ^{111}Cd, and ^{71}Ga. The differences in the peak intensities of these 7 elements between the two groups were significant with a 99.99 % CI (refer to Table 1). In previous research, the mean concentration of Sr in Chinese pepper samples was higher than those in other regions. It was originated from the geological background of the thin limestone of the Triassic Daye Formation [28]. This suggests that the XGBoost model effectively utilized more sample-discriminant elements for differentiation of the two groups.

Eight elements yielded an intensity difference with a 99.99 % CI (refer to Table 1). Among them, only ^{63}Cu was not selected among them, probably because the elemental information of Cu would be duplicative of the 7 other selected elements. The correlation between the intensities of ^{133}Cs and ^{63}Cu peaks was examined for confirmation. The resulting correlation coefficient was relatively high, 0.73, so the Cu information was redundant and not in high demand for discrimination. The 8th and 10th contributing elements were ^{95}Mo and ^{205}Tl, showing 99.9 and 95.0 % CIs in intensity difference, respectively. The 9th contributor was ^{39}K (yielding the strongest intensity), although the corresponding peak intensity difference between groups was not significant. This implies that the ^{39}K peak is somewhat useful for discrimination due to its high sensitivity.

In the XRF measurement, among the 6 elements (Cu, Zn, Rb, Cl, Br, and Ag) with element concentration differences greater than 95.0 % CI (refer to Table 1), only Br was not among the top 10 contributors (refer to Fig. 2 (b)), indicating that it is not critical for the discrimination. It is interesting that Cu was the 2nd contributor even though it was not included in the top 10 contributors in the LA-ICP-MS measurement. The remaining elements of K, Sn, S, P, and Al minorly contributed to the discrimination.

Overall, in both LA-ICP-MS and XRF measurements, XGBoost utilized the more sample-discriminant elements early on and then included the less discriminant elements in the model to maximize the discrimination accuracies. The different elemental compositions in the imported and domestic chili pepper samples clearly enabled sample differentiation according to geographical origin. As discussed earlier, diverse factors such as soil/water conditions in a cultivation area, genetic origin of a plant, and processing conditions caused the dissimilar elemental compositions [5–8].

Merging element peaks obtained from the LA-ICP-MS and XRF measurements would help to enhance the discrimination accuracy further since the two types of elemental information may be complementary to each other. When the merged element data were used for XGBoost, the accuracy was 96.8 %, only slightly higher than that (96.2 %) based on the use of LA-ICP-MS data. Since the LA-ICP-MS measurement already provides diverse and sample-discriminant element peaks, the addition of XRF data providing a smaller number of sample-discriminant element peaks does not appear to be synergetic in improving accuracy.

3.3. Discrimination of imported and domestic samples using NIR spectroscopy

Since the compositions of organic components in the chili pepper samples could also vary depending on geographic origin, NIR spectroscopy, which is able to highlight molecular signatures of the components, could be a useful choice for discrimination. Fig. 3 (a) shows the average MSC-treated NIR spectra (6050-4000 cm^{-1} range) of imported (red) and domestic (black) chili pepper samples. The shade on each average spectrum corresponds to the maximum variation of absorbance at each wavenumber. The average spectra without shade are also displayed for easier comparison. In both NIR spectra, two strong peaks at 4334 and 4258 cm^{-1}, originated from the combination of CH_3 asymmetric

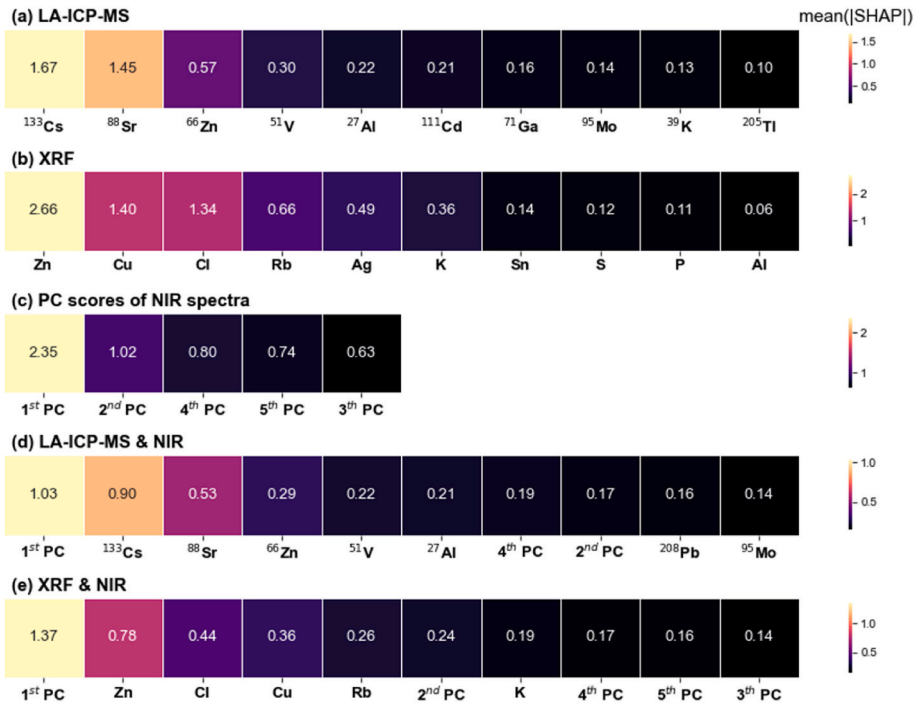

Fig. 2. The input variables yielding the highest mean absolute SHAP values in the discrimination of imported and domestic chili pepper samples in LA-ICP-MS (a), XRF (b), NIR (c), combined LA-ICP-MS & NIR (d), and combined XRF & NIR (e) measurements. The corresponding variables and their mean absolute SHAP values are indicated below and inside each box, respectively. The variables in each measurement are arranged in descending order of mean absolute SHAP values.

stretching and CH_2 bending and the combination of CH_3 symmetric stretching and CH_2 bending of aliphatic structures, respectively [29], were apparent. Also, two broad and weak overtone peaks were observed in the 5900-5600 cm^{-1} range [30]. The peaks in the 5000-4450 cm^{-1} range originated from combination vibrations of N–H and C=O bonds (designated as amide I and amide II bands), and broad peaks around 5200 cm^{-1} are from the combination of OH bonds as well as the first overtone of OH bonds [30].

To examine the spectral features in detail, the spectra of main constituents in chili pepper (including carbohydrates, fatty acids, capsaicin, amino acids, and vitamins) were examined as shown in Fig. 3 (b). The spectra are offset for comparison. First, peaks at 5800, 5680, 4334, and 4258 cm^{-1} are apparent in the spectrum of linoleic acid. So, these peaks in the sample spectra are from components with long aliphatic hydrocarbon chains such as fatty acids and lipids. The spectral features of retinol and cellulose are relatively broad and simple, such as the peaks at 5200 and 4700 cm^{-1}. This suggests that the components containing numerous hydroxyl groups are attributed to these two peaks in the sample spectra. The sum of linoleic acid and cellulose spectra is quite similar to those of the chili pepper samples. It indicates that carbohydrates and lipids are the major components contributing to the NIR spectra, as in Fig. 3 (a). In these spectra, the intensities of carbohydrate and lipid peaks are greater for the domestic and imported samples, respectively. This observation is accordance with the findings in previous studies, in which the higher carbohydrate and lower lipid contents in the domestic samples than the imported samples were reported by the compositional analysis [31,32].

Capsaicin has several bands in the 4700-4000 cm^{-1} (combination band) range and a peak at 5900 cm^{-1}. For thiamin and ascorbic acid, individually characteristic peaks appeared in the combination band range. Also, peaks appeared at 5140 and 5080 cm^{-1} for thiamin, and the peak at 5700 cm^{-1} for ascorbic acid was rather distinctive. For 6 amino acids (aspartic acid, glutamic acid, isoleucine, leucine, proline, and tryptophan), different peaks were apparent in the combination band range as well as in the 6100-5600 cm^{-1} (overtone band) range. Meanwhile, the spectral features of isoleucine and leucine were analogous to each other due to their similar molecular structure, and the peaks of tryptophan were relatively distinct to those of the remaining amino acids due to the possessed indole side chain.

Based on the pure component spectra, the major nutrients of carbohydrates, lipids, amino acids (proteins), capsaicin, and vitamins contribute to the NIR spectra of chili pepper samples. In both average spectra, the intensities of peaks in the 5900–5350 and 4400-4100 cm^{-1} ranges were higher for the imported samples, while they were lower in the 5240-4490 cm^{-1} range compared to those of domestic samples. The differences originated from the dissimilar organic composition of samples between the two groups are valuable sources enabling discrimination. Since the shades of the two average spectra overlap, clear two-group discrimination is unrealizable.

For the discriminant analysis, PCA was performed using all the sample spectra for dimension reduction, and the first 5 principal component (PC) scores were fed for *k*-NN, SVM, PLS-DA, and XGBoost. These PCs described 99.1 % of the total spectral variation in the whole sample set (1st PC: 70.17 %, 2nd PC: 24.53 %, 3rd PC: 3.20 %, 4th PC: 0.94 %, 5th PC: 0.30 %). For reference, the distributions of the 1st and 2nd PC scores of the samples are presented in Fig. S2 (Supplementary Information). The accuracies were 93.5, 92.8, 92.7, and 94.1 % when *k*-NN, SVM, PLS-DA, and XGBoost were employed, respectively (refer to Table 2), with XGBoost yielding the highest accuracy. This suggests that XGBoost is more effective for extracting necessary information from the PC scores for two-group differentiation. XGBoost generates each tree with considering feature importance of input variables. Since it incorporates the important feature into decision trees for discrimination, the influence of unnecessary features could be reduced and the important features could be more weighted in a model, as described in former reports [16,33]. It explains the improved discrimination accuracy.

Nonetheless, the achieved accuracy is inferior to that using LA-ICP-MS data (96.2 %). Fig. 2 (c) shows the mean absolute SHAP values of the 5 PC scores. As expected, the SHAP value of the 1st PC score was highest since it describes the largest spectral variation in the dataset, and then the 2nd PC score yields the second highest SHAP value. In the cases of 3rd, 4th, and 5th PC scores, the corresponding SHAP values were not

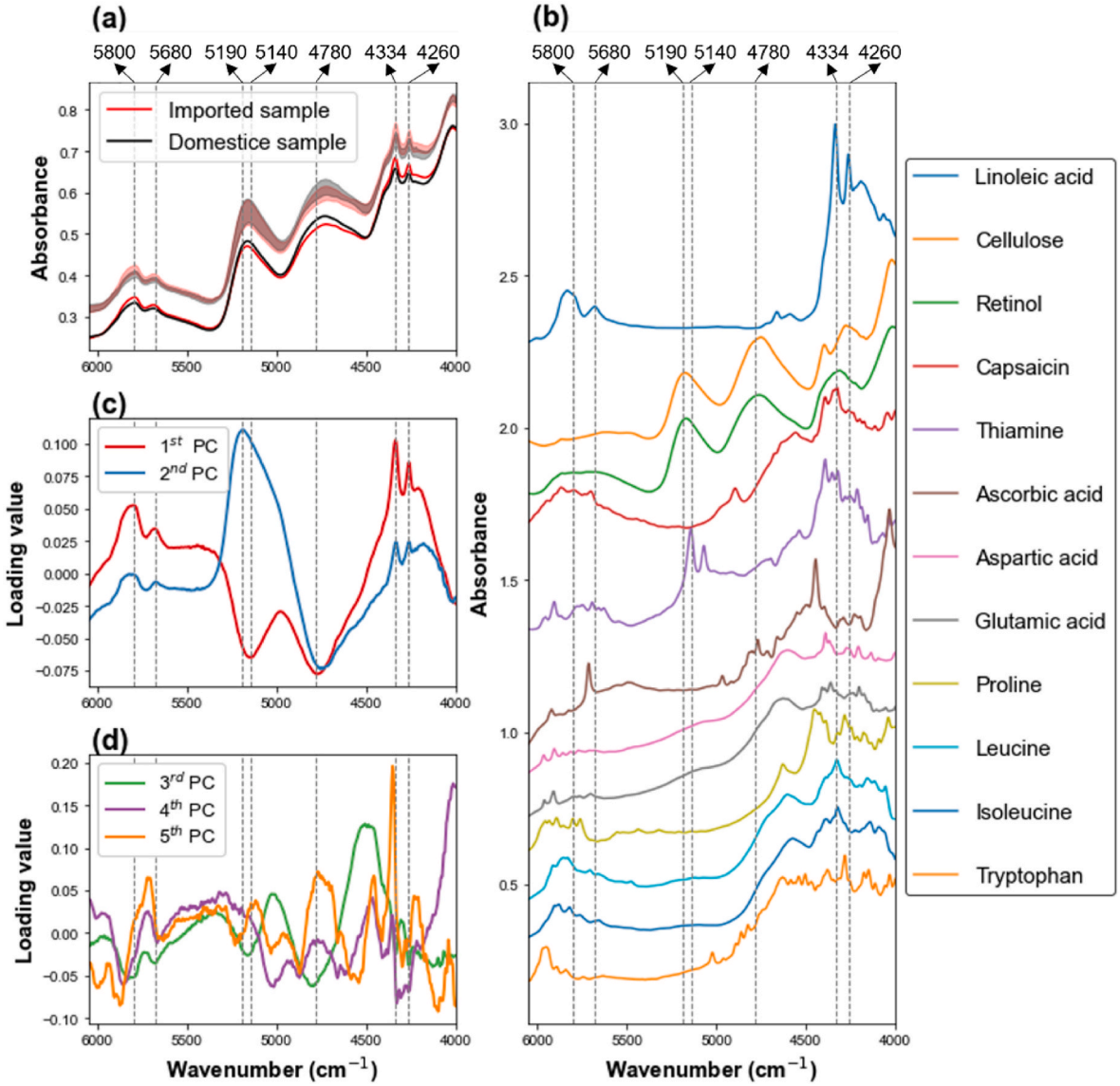

Fig. 3. Average NIR spectra of imported (red) and domestic (black) chili pepper samples (a) in the 6050-4000 cm^{-1} range (spectra at the top). The shade on each average spectrum corresponds to the maximum variation of absorbance at each wavenumber. The average spectra without the shades are additionally presented at the bottom for easier comparison. NIR spectra of the 12 pure components (b) and the 1st/2nd (c) and 3rd/4th/5th PC loadings (d) obtained from the NIR spectra of all the chili pepper samples are also displayed. The vertical dotted lines from the left in each plot indicate the wavenumbers at 5800, 5680, 5190, 5140, 4780, 4334, 4260 cm^{-1} where the weights of 1st and 2nd PC loadings are high. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

in the same order; for example, the 3rd PC score is the 5th contributor. By the way, the SHAP values of these 3 PC scores were small and not significantly different from each other.

Fig. 3 (c) and (d) show the loadings of the 5 PCs. The 1st PC loading highlights all the major bands in the sample spectra, and the 2nd PC loading places more weight on the 5200 cm^{-1} peak (Fig. 3 (c)). The 3rd PC, 4th PC, and 5th PC loadings explain the diverse localized peaks (Fig. 3 (d)), while their percentages of spectral description to the total spectral variation were small, all below 3.20 %. This suggests that the 3rd PC, 4th PC, and 5th PCs contributed little to the discrimination.

3.4. Combination of LA-ICP-MS or XRF data with NIR data for discrimination

The combination of elemental data (obtained by LA-ICP-MS or XRF) and NIR data of organic constituents could be a potential way to enhance the discrimination further since the two intrinsically different spectral features would be complementary and synergetic to differentiate the samples according to their geographical origins. For example, the combination of NIR and laser-induced breakdown spectroscopy (LIBS) was advantageous to enhance the accuracies for identification of geographical origins of milk vetch root (*Astragalus membranaceus*) and soybean paste samples [34,35]. For evaluation, the 5 PC scores of NIR spectra were combined with the intensities of element peaks acquired by LA-ICP-MS or XRF, and the merged data in each combination were input for XGBoost. The results are summarized in Table 2.

The accuracy was improved to 97.5 % by combining the XRF and NIR data (90.5 and 94.1 %, respectively, when separately used). This confirms that the information on compositions of elements and organic constituents in the samples was complementary for discrimination. Also, the merged LA-ICP-MS and NIR data resulted in improved accuracy of 97.2 %, though it was slightly lower than that (97.5 %) using the merged XRF & NIR data. Fig. 2 (d) and (e) present the variables yielding the top 10 highest mean absolute SHAP values in the XGBoost model using the

LA-ICP-MS & NIR and XRF & NIR merged data, respectively. In both cases, the NIR 1st PC score most significantly contributed (the highest SHAP value) to the discrimination. When using the LA-ICP-MS & NIR merged data, Cs, Sr, Zn, V, and Al were the primary contributors. The intensities of these 5 elements between the two groups were different with 99.99 % CI (refer to Table 1). Next, the 4th PC score, 2nd PC score, Pb, and Mo were the following contributors. With using the XRF & NIR merged data, Zn, Cl, Cu, and Rb are the following contributors after the 1st PC score. These four elements, the top four contributors in the XRF measurement, are chosen again. Next, the remained four NIR PC scores and K are the 6th to 10th contributors. The most intense K peak was again informative for the discrimination, thereby chosen as the 7th contributor. For both cases of merged data, XGBoost effectively utilized the complementary elemental peak intensities and NIR PC scores for discrimination, and the subsequent accuracy was thereby enhanced.

In the case of the LA-ICP-MS & NIR data, the sum of the SHAP values for the NIR and LA-ICP-MS was 1.58 and 4.07, respectively, and their ratio was 1:2.58. This means that the variables in the LA-ICP-MS data were more important and influential for the classification. However, in the case of XRF & NIR data, the sums of the SHAP values of each data were quite similar to each other, such as 2.08 for NIR data, and 2.51 for XRF data (a ratio of 1:1.21). This implies that the two different data are rather complementary to each other for the classification model. As a result, the subsequent accuracy was also improved.

Fig. 4 (a) shows the SHAP values for the selected 10 variables of all the samples instead of their mean absolute SHAP values in the case of employing XRF & NIR merged data. The *x*-axis designates the SHAP values of the samples. As described earlier, positive and negative SHAP values imply contributions on the differentiation into imported and domestic samples, respectively. The colors on the symbols indicate the magnitudes of corresponding input variables, red and blue for higher and lower values, respectively. As expected, the SHAP values for the 1st PC score of the samples varied most widely (to either the left or right). This implies that the 1st PC score had the greatest contribution (larger SHAP value) to the discrimination, and the magnitudes of the 1st PC scores were greater for the imported samples (larger number of red-colored symbols at the far right). For a more detailed examination, the SHAP values for the 1st PC scores and the 1st PC score values of the samples were plotted together in Fig. 4 (b). The red and black circles correspond to the imported and domestic samples, respectively. The 1st PC score values of imported and domestic samples were mostly greater and smaller than the horizontal border of the PC score of zero, respectively (refer to the *y*-axis scale). Also, large numbers of imported and domestic samples were densely located at SHAP values around 1.6 and −1.5, respectively, indicating strong positive and negative contributions for the two-group discrimination. Only a small portion of the sample was located in between the two ends. This observation demonstrates that the 1st PC score is a critical variable in the authentication of chili pepper samples.

The observation of the 1st PC loading in Fig. 3 (c) could give a clue to justify the 1st PC score as the 1st contributor. As seen in the figure, the loading values of the 1st PC are negative for the carbohydrate peaks at 5200 and 4700 cm^{-1}, and positive for the lipid peaks at 4334 and 4258 cm^{-1}. Carbohydrates and lipids were the main constituents in chili pepper as described earlier (refer to Fig. 3), and the contents of carbohydrates and lipids in the domestic samples were higher and lower, respectively, than the imported samples as reported by the former compositional analysis [31,32]. Therefore, the 1st PC loading effectively weighted the corresponding NIR peaks to recognize the differed contents of carbohydrates and lipids. This is why the 1st PC score became the 1st contributor to the two-group discrimination.

In the case of Zn (Fig. 4 (a)), the 2nd contributor, the SHAP value variation was smaller than that of the 1st PC score, which corresponds to a lower contribution to the discrimination. Again, large numbers of imported and domestic samples were again considerably populated at the left and right ends, supporting an important role of Zn in the two-group differentiation. Fig. 4 (c) shows the distribution of SHAP values for Zn vs. Zn concentration of the samples. The Zn concentrations of imported and domestic samples were generally smaller and greater than the horizontal border of Zn concentration at 25, respectively (refer to the *y*-axis scale). The imported (red) samples were located over not only the positive SHAP values, but also the negative SHAP values. In contrast, the domestic (black) samples were almost populated over the negative SHAP values. It suggests that the concentration of Zn is an important variable in the authentication model.

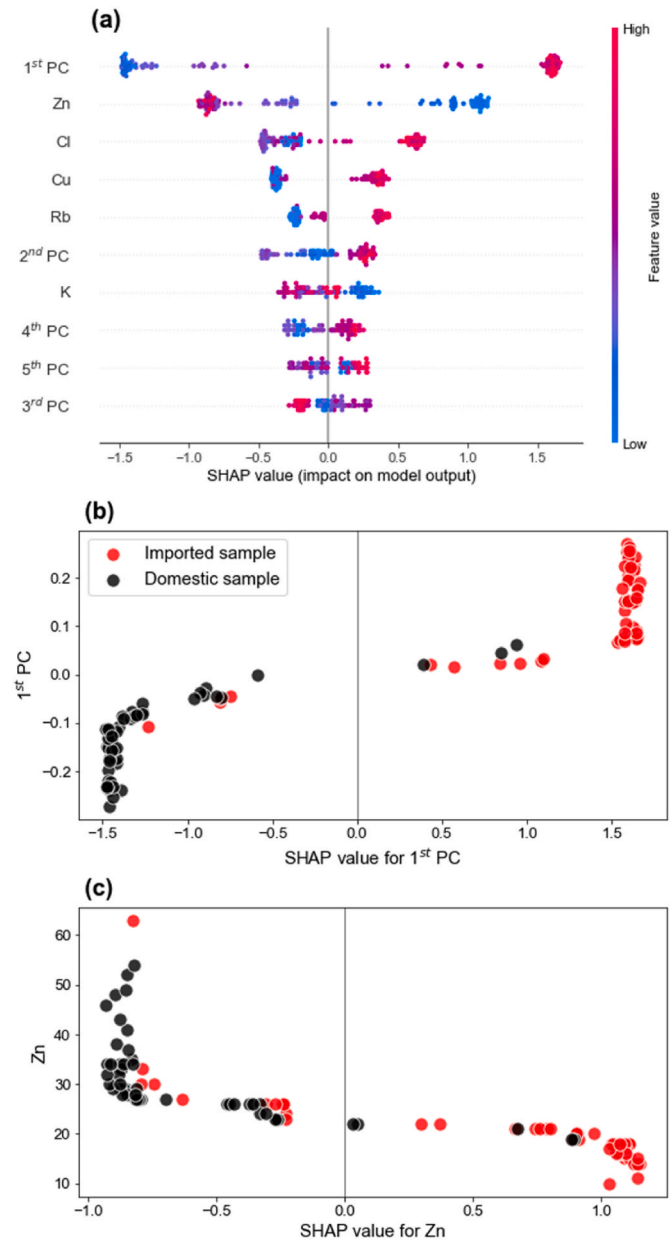

Fig. 4. SHAP values corresponding to the top 10 variables of all the samples in the case of employing XRF & NIR merged data (a). The *x*-axis designates the SHAP values of the samples. The colors on the symbols indicate the magnitudes of corresponding input variables, red and blue for higher and lower values, respectively. Distribution of SHAP values for 1st PC score vs. 1st PC score values (b), and Zn vs. Zn concentrations (c) are also shown. The more positive and negative SHAP values imply more contributions on the differentiation into imported and domestic samples, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

In a former publication [36], the negative correlation between the soil pH and adsorption of Zn was reported. Moreover, when the pH and Zn concentration in the soils in China and Korea were compared, the pH values of Chinese soils were greater than those of Korean soils. Therefore, subsequent Zn concentrations in the soils in both countries would be dissimilar to each other. Actually, the high Zn concentration in Korean soils was reported in parallel [37]. The findings in these two reports support that the contribution of Zn concentration (high SHAP value for Zn) in the authentication of chili pepper samples, thereby becoming the 2nd contributor to the two-group discrimination.

After Zn, the spans of SHAP value variation become smaller in the order of Cl, Cu, and Rb (3rd, 4th, and 5th contributors, respectively), while the localizations of samples at the ends are moderate (Fig. 4 (a)). This suggests that the concentrations of Cl, Cu, and Rb moderately contribute to discrimination. For the remaining 5 variables (6th to 10th contributors: 2nd PC score, K, 4th PC score, 5th PC score, 3rd PC score), the spans of SHAP value variations were not narrowed but more populated in the middle, indicating degraded discrimination ability of these variables. For reference, plots showing the distribution of the SHAP values vs. magnitude of variable for the remaining 8 variables (3rd to 10th contributors) are provided in Fig. S3 (Supplementary Information).

It would be beneficial to search the roles of Cl, Cu, and K (3rd, 4th, and 7th contributors, respectively) for the discrimination, while they are hardly explainable at this stage. Nonetheless, it is reasonable to postulate that multiple factors such as soil/water conditions in a cultivation area, genetic origin of a plant, and processing conditions are associated with the dissimilar elemental compositions [5–8]. In the cases of 2nd, 4th, 5th, and 3rd PC scores (6th, 8th, 9th, and 10th contributors), the corresponding loadings describe the spectral features related to the minor organic constituents such as capsaicin, amino acids, and vitamins, along with the major constituents of carbohydrate and lipid. Therefore, they were within the top 10 contributors. Overall, examination of the SHAP values of the 10 selected variables is beneficial to understand how each feature contributes to discrimination of imported and domestic chili pepper samples in the XGBoost model.

4. Conclusions

The use of XGBoost improved the discrimination accuracies for all three measurements (LA-ICP-MS, XRF, and NIR) compared to those of *k*-NN, SVM, and PLS-DA. This suggests that XGBoost effectively utilizes the input variables for target analysis, the discrimination of imported and domestic chili pepper samples. Moreover, when the LA-ICP-MS or XRF and NIR data were merged for XGBoost, the accuracies reached 97.2 and 97.5 %, respectively. The accuracy improvement was attributed to the combination of complementary atomic and molecular spectroscopic signatures of the samples. Additionally, it was valuable to examine how atomic and molecular information of the samples contribute to the discrimination model through the corresponding SHAP values. This proposed explainable machine learning method helped overcome the black-box nature of the existing XGBoost model and is worth expanding to further applications. From a practical viewpoint, the prices of XRF and NIR instruments are relatively affordable, and corresponding portable instruments are widely available. Therefore, on-site authentication of agricultural samples could be viable using the demonstrated scheme combining XRF and NIR data for XGBoost-based discriminant analysis. This task is under way in our research group.

Meanwhile, in the case of using XGBoost for an intended analysis, there is a possibility for overfitting of a model when the numbers of samples (analytical data) are few or limited. Since XGBoost is a method of continuous learning to reduce residuals, a model likely trains also irrelevant information or noise for a target analysis under this circumstance [33] and a subsequent result would be erroneous as well as lack of robustness. Numbers of samples subject to spectroscopic analysis are often not large, so a precaution is necessary when employing XGBoost as a machine learning tool. Also, beyond SHAP values, a method able to provide more detailed and descriptive contributions, such as important spectral regions, will be helpful for explainable XGBoost modeling. This is an issue under research in this group.

CRediT authorship contribution statement

Seongsoo Jeong: Formal analysis, Data curation, Conceptualization. **Yong-kyoung Kim:** Investigation, Conceptualization. **Suel Hye Hur:** Formal analysis. **Hyojoo Bang:** Formal analysis. **HoJin Kim:** Supervision, Conceptualization. **Hoeil Chung:** Writing – review & editing, Writing – original draft, Supervision.

Declaration of competing interest

To the best of our knowledge, the named authors have no conflict of interest, financial or otherwise.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jafr.2024.101446.

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Abbreviation

LA-ICP-MS: laser ablation-inductively coupled plasma mass spectrometry
XRF: X-ray fluorescence
NIR: near-infrared
XGBoost: eXtreme Gradient Boosting
SHAP: SHaply Additive exPlanation